

Text preprocessing

- Representations for Financial Text -

Department of Data Science

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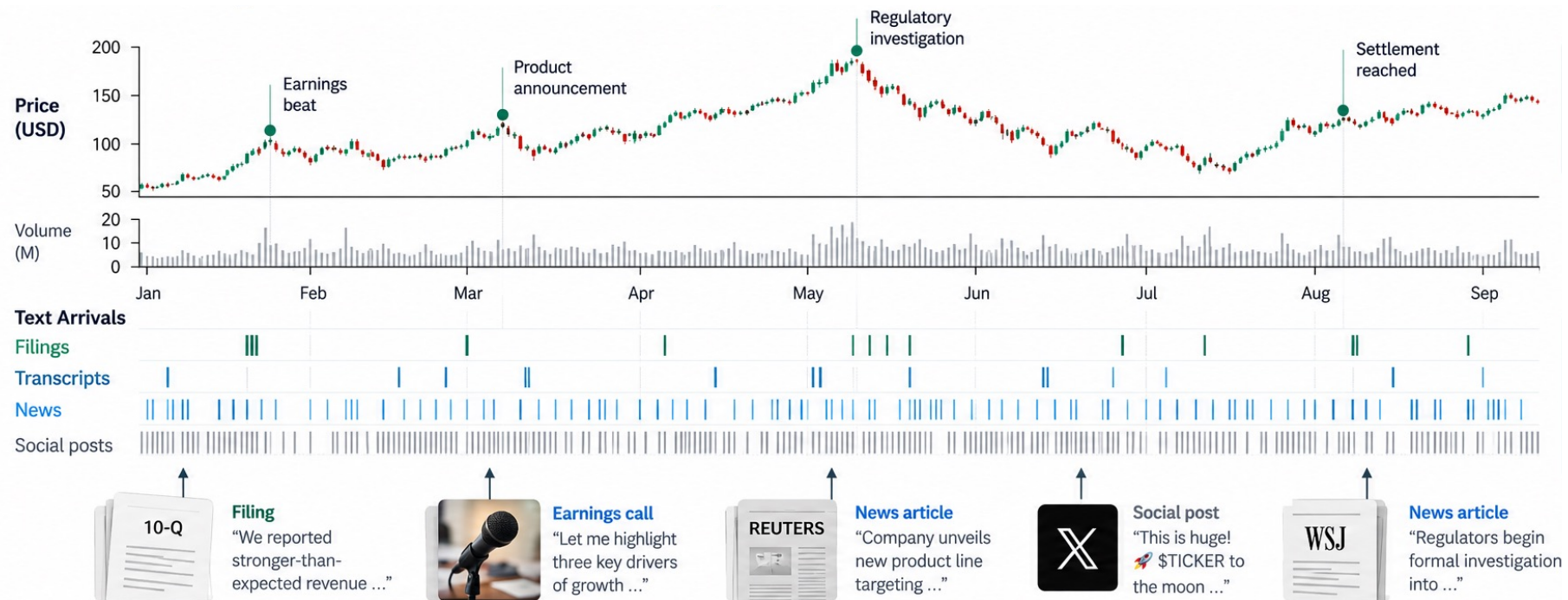
이화여자대학교
EWA WOMANS UNIVERSITY



The Incompleteness of a Price-Only View

Every feature so far came from a price or a volume.

- A price records that the market moved. It does not record why it moved.
- The reasons are written down as text: filings, call transcripts, news copy and forum postings.
- Most of it is free, legal and available at scale, so the barrier is processing.



What Arrives About a Company

Four channels carry text about a company, and each has its own cost structure.

Source	Character	Timing	Access
EDGAR filings (10-K, 10-Q, 8-K)	Long, formal, structured	To the minute	Free, public
Earnings call transcripts	Spoken, then unscripted	Same evening	Commercial
News wire copy	Edited, factual	Minutes to hours	Mostly licensed
Social and forum posts	Noisy, enormous volume	Fastest, least reliable	Partly free

- Row one is filed under a legal obligation. The other three are written for readers, not for models.

Row 1 and part of row 4 are the rows a student can reach for free.



Alternative Data

Sources outside the financial statements, used because they move before the statements do.

Data source	Claimed to measure	How seriously taken
Satellite parking lot imagery	Store traffic before earnings	Commercial, peer-reviewed
Credit card panels	Consumer spending	Widely bought
Job postings	Hiring and expansion	Established in academic work
Shipping and port activity	Trade volume	Evidence at index level
News, filings text, social posts	Attention, tone, disagreement	The subject of this course

Text has the widest coverage and the lowest cost, and the worst signal-to-noise ratio.



The Pentagon Pizza Index

- **The story:** late pizza orders near the Pentagon are said to rise before a major announcement.
- **The appeal:** it needs no privileged access, and nobody involved intends to disclose anything.
- **The evidence:** news stories and social threads only,
with no published series to test.

<https://www.pizzint.watch/>

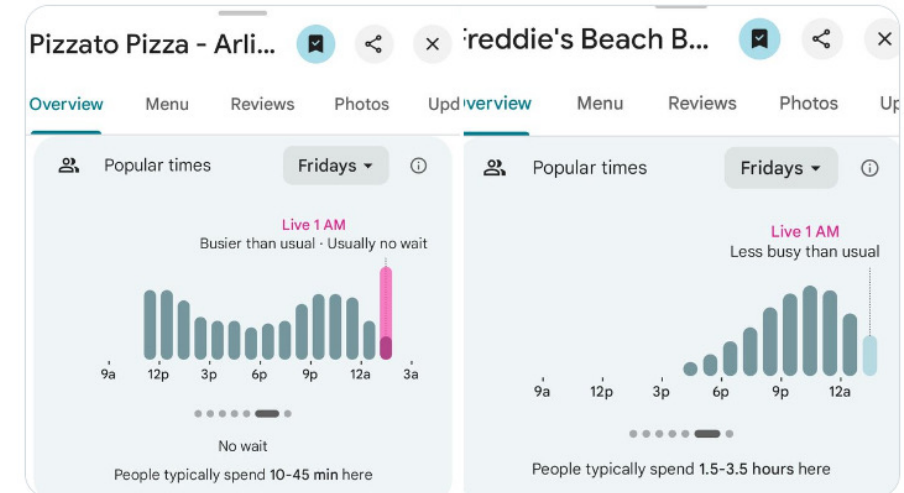


The nearest open pizzeria to the Pentagon, Pizzato Pizza, is reporting high traffic.

Freddie's Beach Bar, the closest gay bar to the Pentagon is reporting very low traffic.

As of 1:28pm ET

[계시물 번역하기](#)



Three Problems With the Pizza Story

Three questions decide whether a pattern is a finding or a story.

The question nobody answered	Why it settles the matter
How many quiet nights also had a spike?	Fridays, paydays and home games spike too. Nobody counted them.
Did anyone write the rule down and then wait?	Every case was collected after the fact. The rule never predicted.
How many other indexes were tried?	Failed attempts are not published, so only the ones that appear to work are visible.

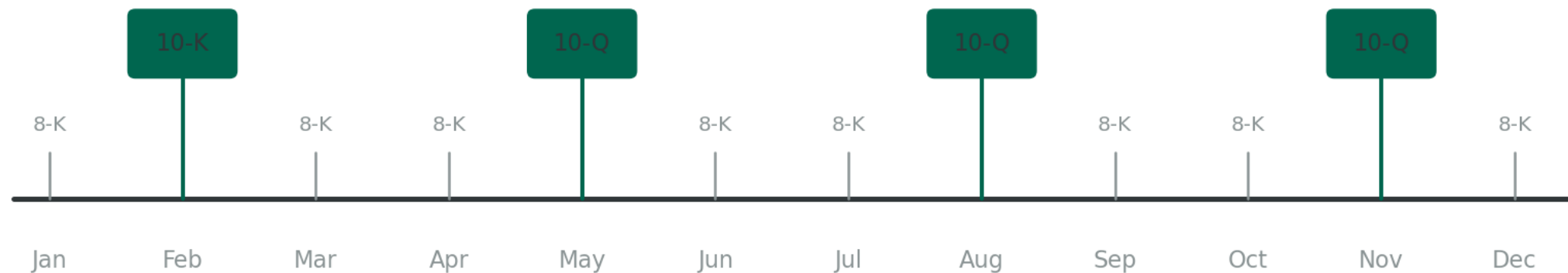
Ask these three questions of every feature you build, including your own.



Mandatory Disclosure as a Text Source

A listed company must publish certain documents on a schedule, in a fixed format.

- In the US these are the 10-K, the 10-Q and the 8-K, filed to EDGAR and free to download.
- They are the cleanest text a company produces: dated to the minute, in a numbered structure.
- This course uses news and forum text, so disclosure appears here as one representative source.



A Page From an Actual 10-K

<https://www.sec.gov/search-filings>



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Exchange Commission

Newsroom | Investors | Small Businesses | Whistleblowers

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Standard Industrial Classification (SIC) Code List

CIK Lookup

EDGAR Public Dissemination Service (PDS) System

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Company Search

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A Page From an Actual 10-K

- Item 1A as it appears in the filing, before any parsing.

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

OR

SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

Commission file number: 0-23985



NVIDIA
NVIDIA CORPORATION
(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction of
incorporation or organization)

2788 San Tomas Expressway, Santa Clara, California
(Address of principal executive offices)

Registrant's telephone number, including area code: (408) 486-2000

Securities registered pursuant to Section 12(b) of the Act:

Trading Symbol(s)
NVDA

Securities registered pursuant to Section 12(g) of the Act:
None

Class
value per share

[Item 1.](#)
[Item 1A.](#)
[Item 1B.](#)
[Item 1C.](#)
[Item 2.](#)
[Item 3.](#)
[Item 4.](#)

[Item 5.](#)
[Item 6.](#)
[Item 7.](#)
[Item 7A.](#)
[Item 8.](#)
[Item 9.](#)
[Item 9A.](#)

[Part I](#)

[Business](#)
[Risk Factors](#)
[Unresolved Staff Comments](#)
[Cybersecurity](#)
[Properties](#)
[Legal Proceedings](#)
[Mine Safety Disclosures](#)

[Part II](#)

[Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities](#)
[\[Reserved\]](#)
[Management's Discussion and Analysis of Financial Condition and Results of Operations](#)
[Quantitative and Qualitative Disclosures About Market Risk](#)
[Financial Statements and Supplementary Data](#)
[Changes in and Disagreements with Accountants on Accounting and Financial Disclosure](#)
[Controls and Procedures](#)



Korean Disclosure Through DART

The Financial Supervisory Service runs DART, the Korean counterpart to EDGAR.

Korean filing	Closest US form	Cadence
사업보고서 (annual report)	10-K	Within 90 days of year end
분기보고서 / 반기보고서	10-Q	Quarterly and semiannual
주요사항보고서	8-K	On a material event

- An OpenDART API serves the filings, but English versions are partial, so a Korean project needs a Korean tokenizer.

<https://dart.fss.or.kr/>



Korean Disclosure Through DART

대한민국 기업정보의 창

DART

ENGLISH >

최근공시

공시서류검색

기업개황

공모정보

OpenDART

공시업무 · 제도

DART 소개

≡



신속하고 투명한 전자공시시스템, DART

공시통합검색

검색조건 : 회사명

회사명/종목코드 입력

Q 검색

최근검색어 선택

☐ 정기공시

☐ 주요사항보고

☐ 발행공시

☐ 지분공시

☐ 기타공시

☐ 외부감사관련

☐ 펀드공시

☐ 자산유동화

☐ 거래소공시

☐ 공정위공시

정기공시 항목별 검색

회사명/종목코드를 입력하세요

Q 검색

회사현황

회사 개요

사업의 내용

경영진단 및 분석의견

감사인의 감사의견

임원 및 직원의 현황

임원의 보수 등

재무정보

요약재무정보

연결재무제표

연결재무제표 주석

개별재무제표

개별재무제표 주석

배당

자금의 조달 및 사용

기타 재무사항

지배구조

최대주주

이사회

감사제도

주주총회

대주주와의 거래

계열회사



Korean Disclosure Through DART

번호	공시대상회사	보고서명	제출인	접수일자	비고
1	 삼성전자	임원·주요주주특정증권등소유상황보고서	허은녕	2026.09.22	
2	 삼성전자	최대주주등소유주식변동신고서	삼성전자	2026.09.18	
3	 삼성전자	임원·주요주주특정증권등소유상황보고서	조혜경	2026.09.18	
4	 삼성전자	임원·주요주주특정증권등소유상황보고서	이혁재	2026.09.18	
5	 삼성전자	임원·주요주주특정증권등소유상황보고서	신제윤	2026.09.17	
6	 삼성전자	임원·주요주주특정증권등소유상황보고서	아포테커마띠유	2026.09.14	
7	 삼성전자	임원·주요주주특정증권등거래계획보고서	이재용	2026.09.09	
8	 삼성전자	임원·주요주주특정증권등소유상황보고서	배상기	2026.09.09	
9	 삼성전자	최대주주등소유주식변동신고서	삼성전자	2026.09.08	
10	 삼성전자	대규모기업집단현황공시(분기별공시(대표회사용))	삼성전자	2026.08.31	
11	 삼성전자	임원·주요주주특정증권등소유상황보고서	김은용	2026.08.31	
12	 삼성전자	주식등의대량보유상황보고서(일반)	삼성물산	2026.08.28	
13	 삼성전자	임원·주요주주특정증권등소유상황보고서	포르치니마우로	2026.08.28	
14	 삼성전자	임원·주요주주특정증권등소유상황보고서	김경륜	2026.08.26	
15	 삼성전자	[기재정정] 임원·주요주주특정증권등소유상황보고서	조미선	2026.08.26	

DART

삼성전자

현재목록

전체문서

검색어 입력

본문

2026.08.14 반기보고서

첨부

+첨부선택+

+작성 정기보고서 바로가기+

영문보기

다운로드

문서목차

반기 보고서

【대표이사 등의 확인】

I. 회사의 개요

1. 회사의 개요

2. 회사의 연혁

3. 자본금 변동사항

4. 주식의 총수 등

5. 정관에 관한 사항

II. 사업의 내용

1. 사업의 개요

2. 주요 제품 및 서비스

3. 원재료 및 생산설비

4. 매출 및 수주상황

5. 위험관리 및 파생거래

6. 주요계약 및 연구개발활동

7. 기타 참고사항

III. 재무에 관한 사항

1. 요약재무정보

2. 연결재무제표

2-1. 연결 재무상태표

2-2. 연결 손익계산서

2-3. 연결 포괄손익계산서

2-4. 연결 자본변동표

2-5. 연결 현금흐름표

3. 연결재무제표 주석

1. 일반적 사항 (연결)

2. 중요한 회계처리방침 (연결)

가. 주식의 총수

2026년 반기말 현재 당사가 발행한 기명식 보통주식(이하 "보통주") 및 무의결권 기명식 우선주식(이하 "우선주")의 수는 각각 7,780,466,850주와 1,194,671,350주입니다. 당사는 2026년 반기말 현재까지 보통주 1,934,188,242주와 우선주 392,300,147주를 이사회 결의에 의거하여 이익으로 소각한 바 있습니다. 2026년 반기말 현재 유통주식수는 보통주 5,764,191,903주, 우선주 802,371,203주입니다.

(기준일 : 2026년 06월 30일)

(단위 : 주, %)

구 분	주식의 종류			비고
	보통주	우선주	합계	
I. 발행할 주식의 총수	20,000,000,000	5,000,000,000	25,000,000,000	-
II. 현재까지 발행한 주식의 총수	7,780,466,850	1,194,671,350	8,975,138,200	-
III. 현재까지 감소한 주식의 총수	1,934,188,242	392,300,147	2,326,488,389	-
	1. 감자	-	-	-
	2. 이익소각	1,934,188,242	392,300,147	자사주소각
	3. 상환주식의 상환	-	-	-
	4. 기타	-	-	-
IV. 발행주식의 총수 (II - III)	5,846,278,608	802,371,203	6,648,649,811	-
V. 자기주식수	82,086,705	-	82,086,705	-
VI. 유통주식수 (IV - V)	5,764,191,903	802,371,203	6,566,563,106	-
VII. 자기주식 보유비율	1.4	0.0	1.2	-

Business Analytics @ Prof. Sangyeop Kim

12

Retail Investor Discussion Boards

Venue	Market	Character
Naver stock boards	Korean equities	Per-ticker, very high volume
Reddit r/wallstreetbets	US equities and options	Heavy irony, attention spikes
StockTwits	US equities	Short posts, bullish or bearish tag

- Read these for attention rather than for opinion: a ticker whose daily post count jumps tenfold usually sees higher volatility.
- Bots post at scale, accounts ramp a position on purpose, and sarcasm inverts the plain words.

차트맛스타

3분 전

즐거운
명절보내세요
전 하닉 어제 꼭대기에 사서 인버스 달달하게
8퍼 수익중입니다
담주 장 시작하면 일주일뒤에 20퍼까만...

👍 0
🗨 1
💬

조회 7

psps2288

21분 전

미중정상회담 하든 삼전닉스에 불리한거 아닌가?
그럴거 같은데...
그나저나 오늘 미장 반도체 떡락이네

👍 1
🗨 5
💬 2

조회 67

u/falling_bac · 2 days ago

Nike down to 2013 levels lmaoo

Meme

👍 8.4K
🗨 374
🔄
🔗 Share

Consensus, Tone and Attention as Measurable Quantities

Quantity	Measured from text as	Used for
Attention	Document count per firm per day	Volume and volatility proxy
Tone	Average polarity of mentions	The sentiment feature
Consensus	Dispersion across authors	Disagreement, not tone

- None of the three needs a language model: counting and averaging already produce a feature.
- All three still need the whole pipeline: collection, cleaning, tokenization and representation.

The aggregation rule affects the result as much as the model that produces the scores.



Counts That Come With the Text

Every platform publishes numbers beside the text, and those numbers are features on their own.

Quantity	Where it comes from	Daily feature	What to watch
Post volume	A board or a subreddit	Posts per ticker per day	Bots and cross-posts
Discussion depth	Comment threads	Comments per post	One argument inflates it
Approval	Upvotes, likes, scores	Median score per post	The score keeps changing
Search interest	Google Trends	Weekly index, 0 to 100	Rescaled per query window
Page views	Wikipedia pageviews API	Views per article per day	Company name collisions

- None of these requires reading the text, and all of them are free at the volumes this course works with.

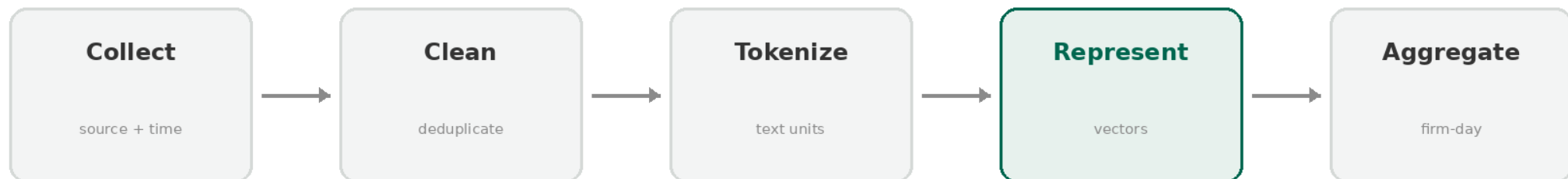
Attention is usually the easiest text feature to build and the first one that works.



The Pipeline

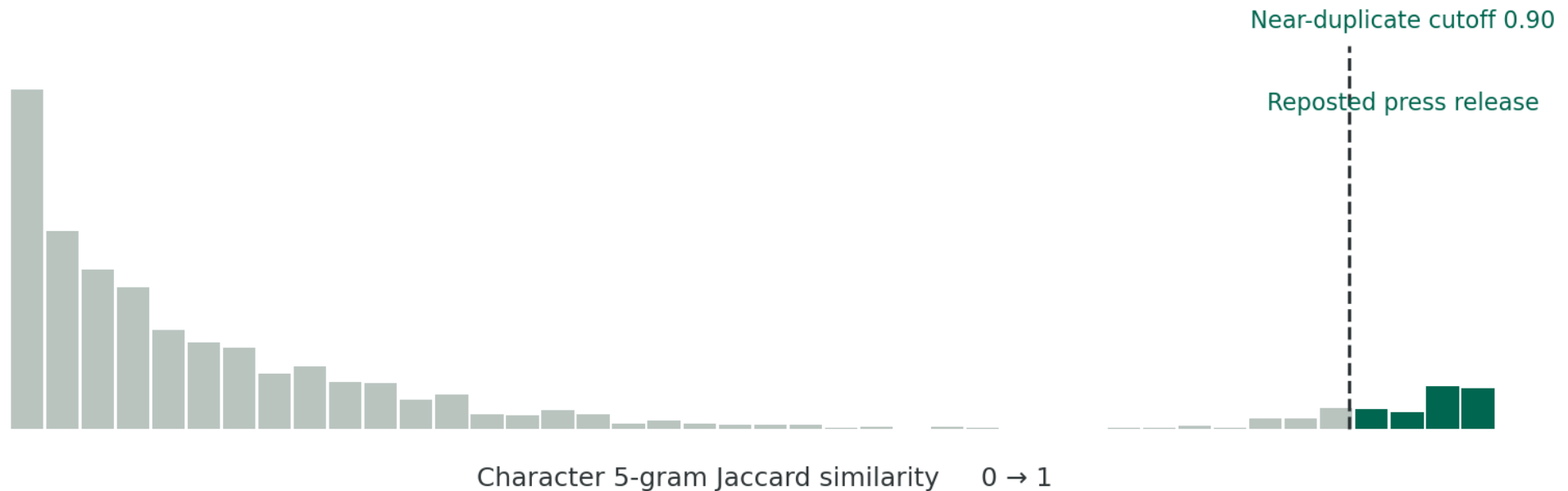
Collect	Clean	Tokenize	Represent	Aggregate
Source, timestamp	Markup, duplicates	Split into units	Vectors	Per day, per firm
A news item pulled at 16:35	Navigation text removed	A long report stays one unit	768 numbers a token	One tone per day

- Stages one to four fix what the model is able to see, before any training starts.



Cleaning

- Markup and boilerplate: strip the HTML tags, the cover page and the safe-harbor paragraph.
- Near-duplicates: compare character 5-gram overlap and cut every pair above a threshold.
- Internal repetition: legal language repeated inside one filing inflates the weight of words that carry no news.



Normalization and Its Traps

Four standard steps, and what each one destroys in a filing.

Common step	Why it is done	Why it hurts here
Lowercasing	Smaller vocabulary	AAPL becomes aapl; U.S. becomes u.s.
Stop-word removal	Drops dull tokens	"no", "not", "without" are on most lists
Punctuation stripping	Cleaner tokens	3.2% becomes 32; \$1.2bn loses its unit
Number removal	Numbers seen as noise	In a filing the numbers are the content

All four are tutorial defaults, and a default is a choice someone else made for you.



Domain Specificity of the Word List

About three-quarters of general negative words found in a 10-K are not negative there.

General "negative" word	What it means in a 10-K
tax	A line item every company has
cost	An accounting category
capital	Capital expenditure, capital structure
liability	The right-hand side of the balance sheet
vice	Vice President

The finance-specific word list comes from Loughran and McDonald, When Is a Liability Not a Liability?, Journal of Finance 66(1).



The Definition of a Token and Its Consequences

A token is whatever unit your model counts.

- Coverage: the vocabulary is finite, so some inputs must be broken into smaller pieces.
- Cost: tokens, not words, fill the context window and decide the size of the API bill.
- Compatibility: two tokenizers cannot share embeddings, counts or vocabularies.

	The company's revenue grew 12% in the third quarter.																						
Word tokenizer (whitespace)	→	The	company's	revenue	grew	12%	in	the	third	quarter	.	10 tokens											
Subword tokenizer (BPE / WordPiece)	→	The	company	's	revenue	grew	12	%	in	the	third	quarter	.	11 tokens									
Character tokenizer	→	T	h	e	c	o	m	p	a	n	y	'	s	r	e	v	e	e	n	u	e	...	44 tokens



Where Whitespace Splitting Breaks on Financial Text

`text.split()` on: "U.S. GAAP requires Apple Inc. to disclose the \$1.2bn charge under Sec. 4.01, up 3.2% from FY2024." gives fifteen tokens and four trailing periods carrying three different meanings.

Construction	Example	What whitespace splitting does
Abbreviation with periods	U.S., Inc., Corp.	U.S. and U.S become different types
Section cross-reference	Sec. 4.01, Item 7A	Splits the label from the number
Money with a scale suffix	\$1.2bn, USD 4.2m	Comma and suffix join the token
Form name with a hyphen	10-K, 8-K, S-1	Dies under punctuation stripping
Percentage, basis points	3.2%, 210 bps	Percent sign never handled consistently
Ticker and identifier	AAPL, CIK 0000320193	Leading zeros vanish if cast

A rule-based fix is an exception list. It is maintainable and useful, but never complete.



Stemming and Lemmatization

Frequency methods split on whitespace, then folds inflected forms together.

Surface form in a filing	Porter stemmer	WordNet lemmatizer
reporting	report	report
reported	report	report
reports	report	report
impairments	impair	impairment
was	wa	be

- Stemming cuts endings by rule, so the output need not be a real word.
- Lemmatization uses a dictionary and a part-of-speech tag, so "was" resolves to "be".

The trade-off is a smaller vocabulary against a vocabulary that stays readable.



Count Vectors on Three Documents

Doc	Text	Tokens
D1	the impairment charge reduced net income	6
D2	the impairment was reversed	4
D3	the company increased its dividend	5

	the	impair	charge	reduc	net	income	was	revers	company	increas	its	divid
D1	1	1	1	1	1	1	0	0	0	0	0	0
D2	1	1	0	0	0	0	1	1	0	0	0	0
D3	1	0	0	0	0	0	0	0	1	1	1	1

36 cells, 15 non-zero. At 30,000 terms the matrix is 99.9% zeros.



TF-IDF: The Two Factors

$$tf - idf(t, d) = tf(t, d) \times idf(t),$$

with tf = count of t in d / length of d and $idf = \ln(N/df(t))$.

Term	df	idf = $\ln(3 / df)$	tf in D1	tf-idf in D1
the	3	$\ln(1) = 0.000$	$1 / 6 = 0.167$	0.000
impairment	2	$\ln(1.5) = 0.405$	$1 / 6 = 0.167$	0.068
charge	1	$\ln(3) = 1.099$	$1 / 6 = 0.167$	0.183

- A term appearing in every document is zeroed, so stop-word removal comes for free.
- The weight rewards a term that is frequent here and rare elsewhere, which is what a distinguishing term is.



Trimming the Vocabulary

Two more settings decide which terms ever enter the matrix.

Setting	What it removes	What it costs in filings
stop_words	The dullest closed-class terms	"no" and "not" go with them
min_df = 5	Terms in fewer than five documents	A rare firm name is real signal
max_df = 0.9	Terms in almost every document	Boilerplate and topic words look alike

- Idf already zeroes a term appearing in every document, so long stop lists add little.
- Set min_df from the corpus size, and report it: it changes the feature space.

Each trimming rule is a modeling choice, not a cleaning detail.



Cosine Similarity: A Worked Example

	Non-zero entries	Length
D1	impairment 0.068; charge, reduced, net, income each 0.183	$\text{root}(0.068^2 + 4 \times 0.183^2) = 0.372$
D2	impairment 0.101; was 0.275; reversed 0.275	$\text{root}(0.101^2 + 2 \times 0.275^2) = 0.402$
D3	company, increased, its, dividend each 0.220	$\text{root}(4 \times 0.220^2) = 0.440$

- D1 against D2: only impairment is shared, $0.068 \times 0.101 = 0.00687$, so the cosine is 0.046.
- D1 against D3: no weighted term is shared at all, so the cosine is exactly 0.000.
- Two sentences describing the same impairment get a similarity near zero.



Linking a News Article to a Company

For each article, find the companies it is really about, out of a list of 500.

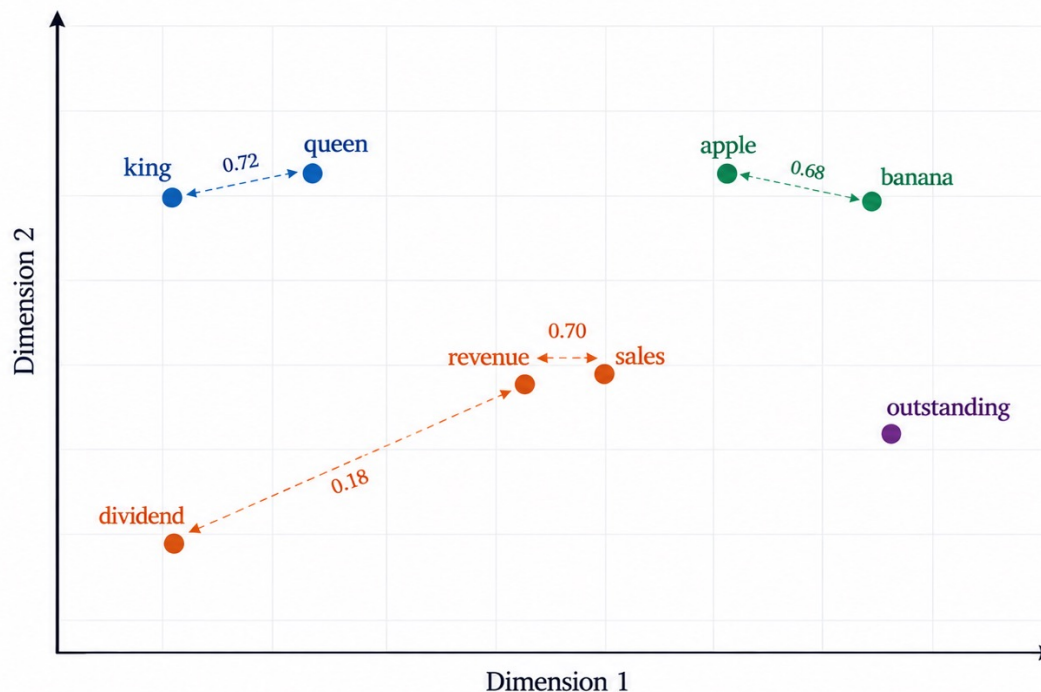
Failure of exact string matching	Example
The name is an ordinary word	"Apple" the fruit; "Target" the verb
Ticker collisions	A cashtag matching an abbreviation
Subsidiaries and brands	Instagram, YouTube, AWS name no listed entity
Legal-name variants	Alphabet Inc., Google LLC, GOOGL
Mentions not about the firm	A sentence naming five competitors

The last row is the hardest case: the name appears, but the article is not about that firm.

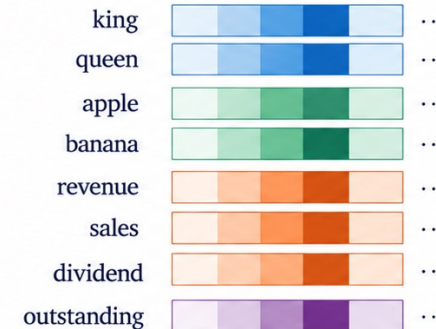


Static Embeddings

- The families: Word2Vec, GloVe and FastText, all trained on plain unlabeled text.
- The object: one vector per word type, about 300 dimensions, learned from neighboring words.
- The gain: "revenue" and "sales" now score about 0.7 against each other instead of zero.
- The limit: the vector is fixed once training ends, so "outstanding" gets one vector for every sense.



Word vectors (300 dimensions)



One vector for all senses

The company reported an
outstanding dividend.

She is an outstanding student.

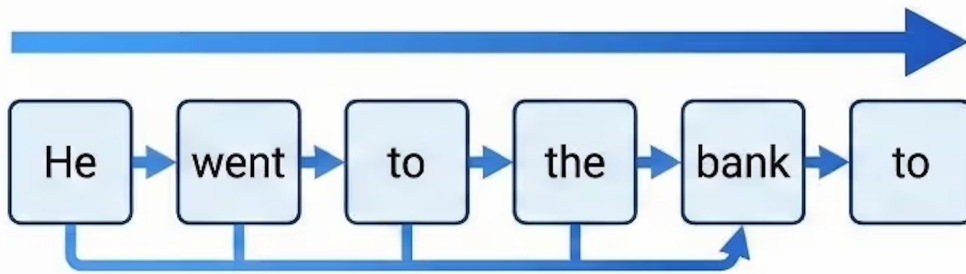


What BERT Is Trained to Do

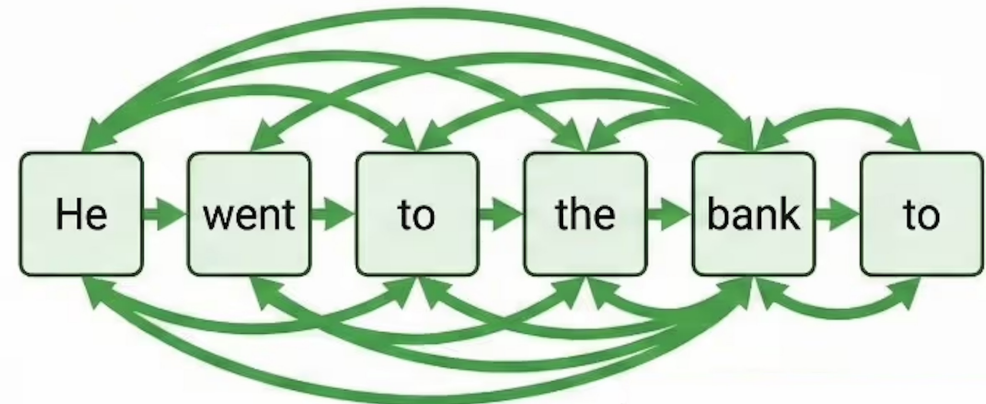
BERT is an encoder trained to fill in deleted words using the words on both sides.

- Keywords: bidirectional, masked language modeling, self-supervised, transfer to a downstream task.
- No human labels are needed, so the training corpus can be the whole of Wikipedia and a book collection.
- Bidirectional matters because a financial word is often disambiguated by the words after it, not only by the words before it.

Traditional Unidirectional Model
(e.g., GPT-1)



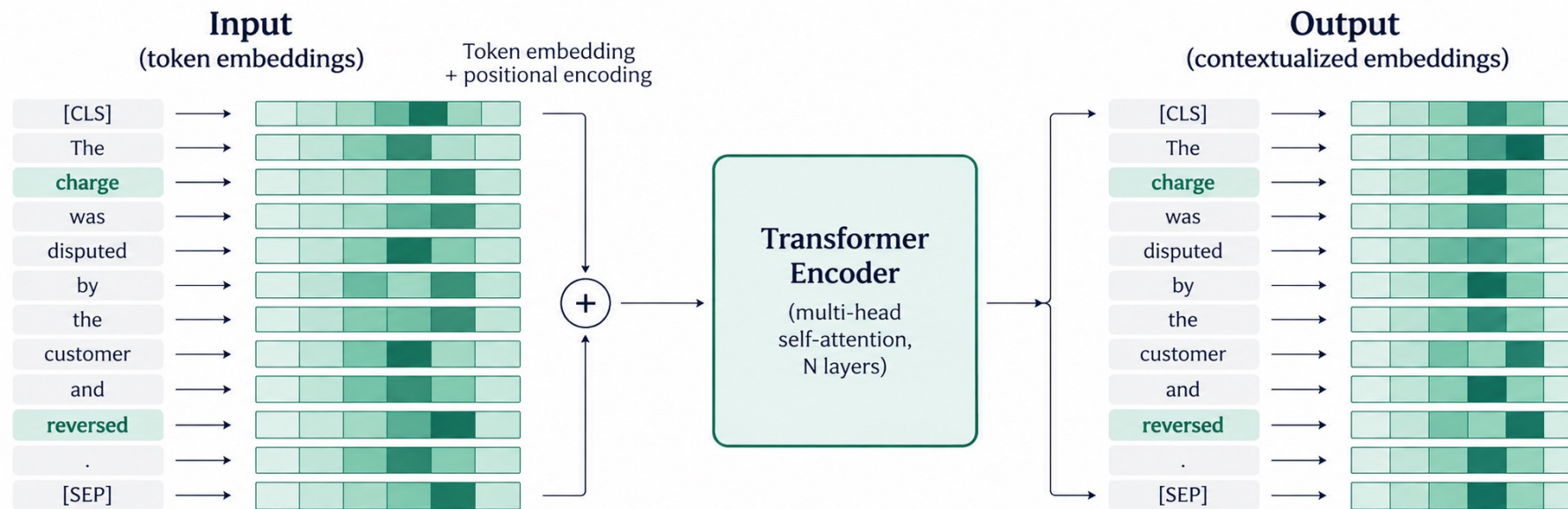
BERT Model
(Bidirectional)



What a Transformer Encoder Takes In and Returns

	What it is	Shape for a 12-token sentence
Input	Integer token ids from a fixed vocabulary	12 integers, plus a 12-entry mask
Output	One real-valued vector per position	12 vectors of 768 numbers

- The shapes match: the number of output vectors always equals the number of input tokens.
- Every output vector is computed from every input position, which is what makes it contextual.



Masked Language Modeling: A Worked Example

Original: The company reported a decline in operating margin this quarter.

Masked: The company reported a [MASK] in operating margin this quarter.

The model's job	Detail
A distribution over the whole vocabulary at the gap	About 30,000 probabilities, summing to one
Use the words on both sides	"reported a" and "in operating margin" both constrain it
Loss against the token that was there	Target decline; rivals drop, fall, reduction

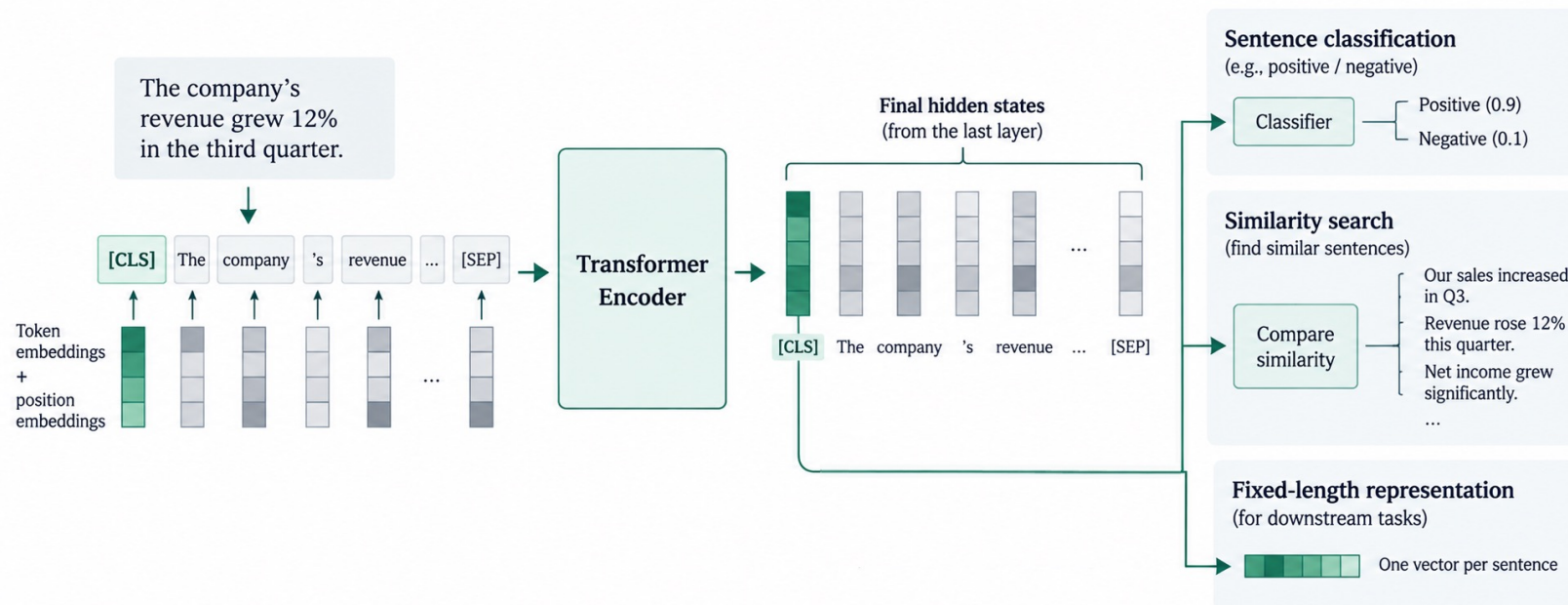
About 15% of positions are masked, and nobody labeled anything by hand.



The [CLS] Token as a Sentence Summary

The tokenizer prepends [CLS], a token with no content of its own, whose output vector is used as the sentence representation.

- It works because pretraining pushed information into it: the next-sentence-prediction objective read the whole pair from that one position.
- Sentiment analysis and every other sentence-level task then attach a small head to that single vector.
- [CLS] is not a good general-purpose embedding on its own: without fine-tuning its cosine similarities are unreliable.



From Token Vectors to a Sentence Vector

The encoder returns one vector per token, and a sentence-level task needs exactly one vector.

Pooling rule	What it takes	When it is used
[CLS] pooling	The vector at position 0	Classification with a fine-tuned head
Mean pooling	Average over positions where the mask is 1	Similarity and retrieval
Max pooling	Elementwise maximum	Rare, keeps only extremes

- Mean pooling must respect the attention mask, or the padding dilutes every short sentence.

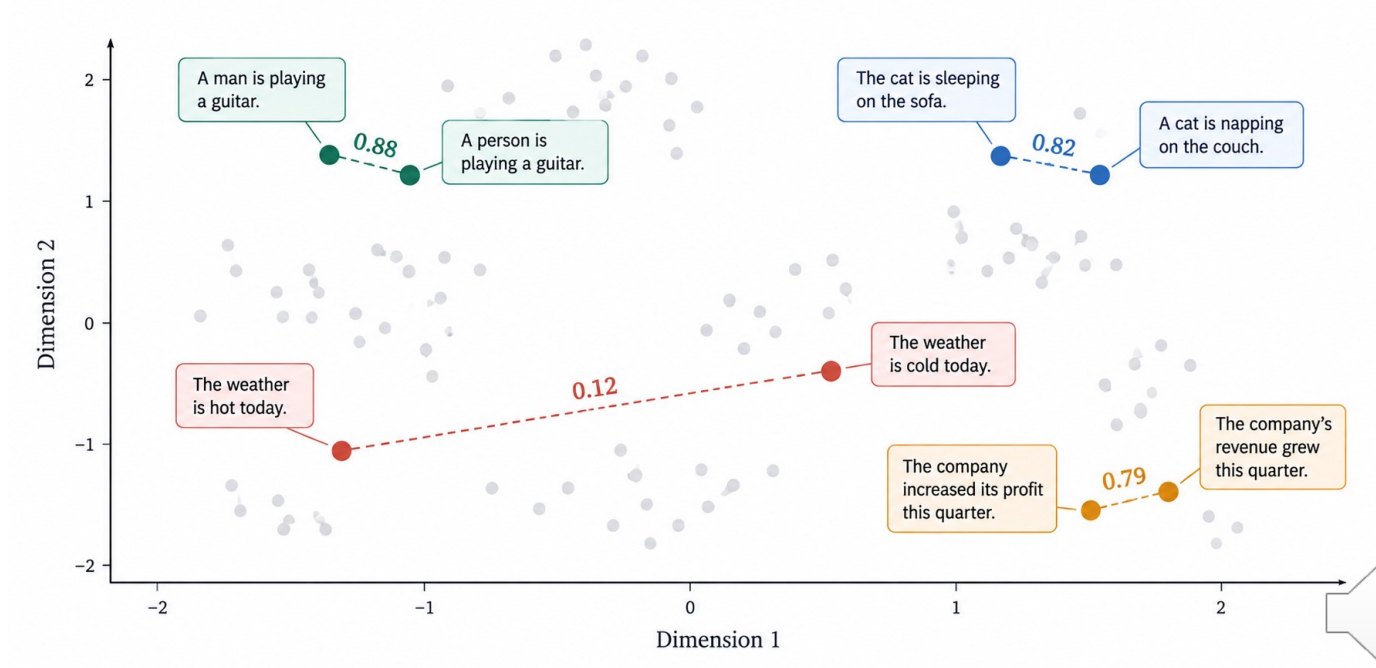


Cosine Similarity in Embedding Space

Once each sentence is one vector, the cosine between two vectors behaves as a similarity of meaning.

Pair	Shared words	TF-IDF cosine	Embedding cosine
Margins narrowed / Profitability weakened	None	0.00	about 0.75
Margins narrowed / Margins improved	Two of three	about 0.60	about 0.25

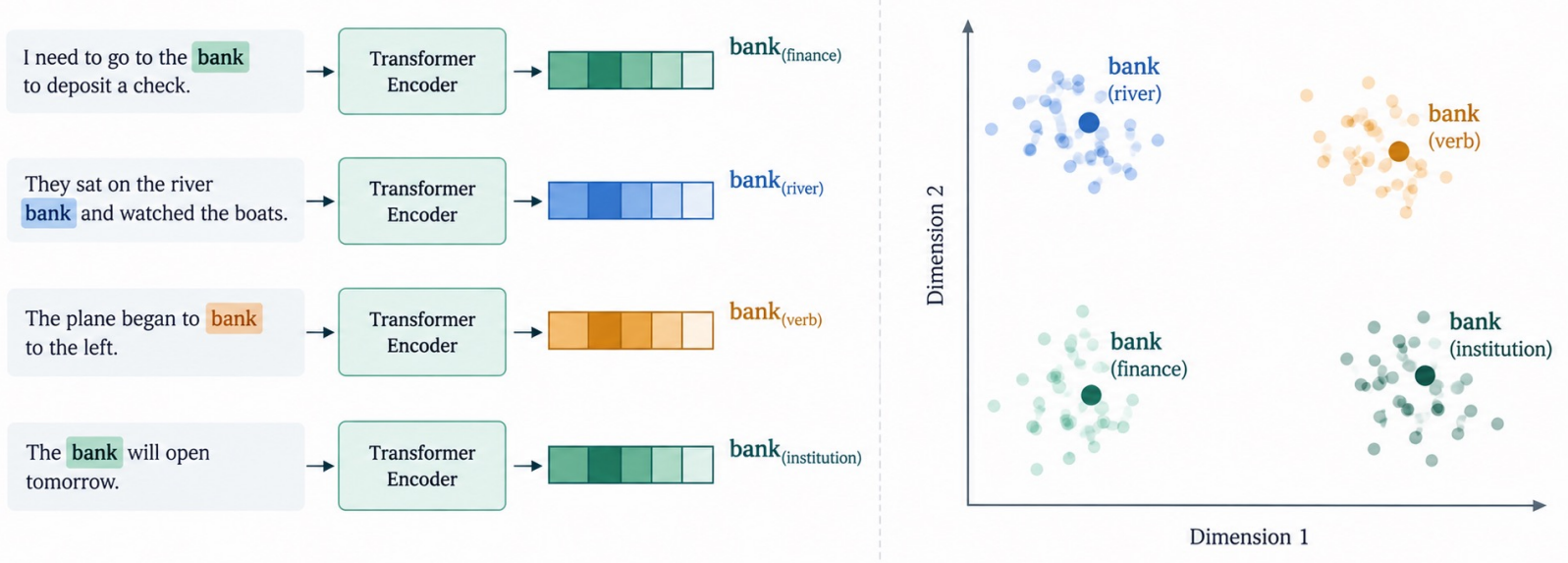
- The second row shows the problem: shared vocabulary and shared meaning are not the same thing.



One Word, Many Vectors

Sentence	What "outstanding" means here
S1: 1.4 billion shares outstanding.	A counting term. Neutral, a balance-sheet item.
S2: Management called the quarter outstanding.	An evaluation. Strongly positive.

A static embedding assigns one vector to the word type. A contextual encoder assigns a vector to the token in its position, so the two sentences separate.



Fine-Tuning Against Feature Extraction

	Feature extraction	Fine-tuning
What is trained	A small model on frozen vectors	Head and all encoder layers
Encoder weights	Unchanged	Updated by the task loss
Labeled examples	Hundreds can be enough	Usually thousands
Compute	One cacheable forward pass	Forward and backward on a GPU
Typical accuracy	Lower, sometimes slightly	Higher when the domain differs

Feature extraction is the standard starting point, and often enough on its own.



The 512 Token Limit

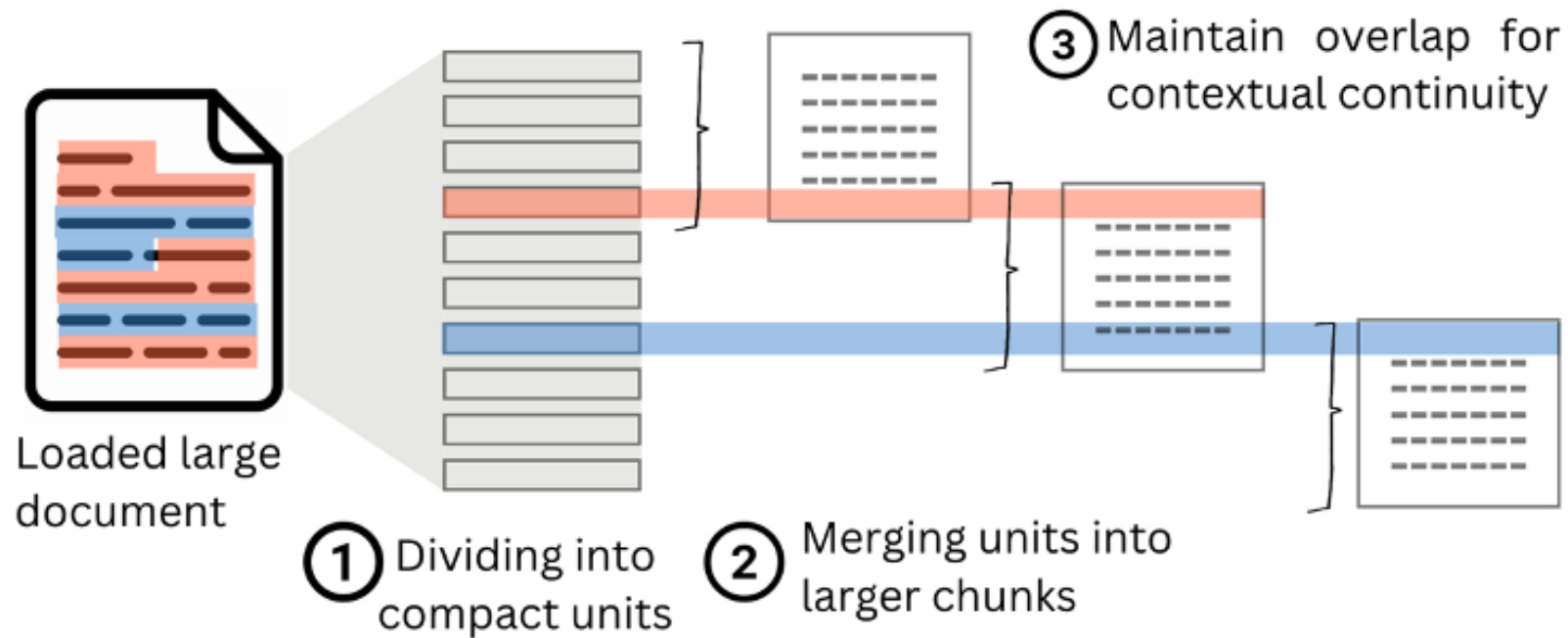
- BERT-base learns one position embedding for each of its 512 positions, so position 513 has no embedding at all.
- The limit is architectural, and `truncation=True` silently discards everything past it.

Document or model	Tokens	Consequence
A news article	300 to 1,500	Fits in one pass
An earnings call transcript	6,000 to 12,000	12 to 24 pieces at 512
An annual report	about 70,000	50,000 words at 1.4 tokens each
FinBERT, BERT-base	512 maximum	137 pieces for one annual report
A frontier model	100,000 and above	A long report fits, at a cost



Chunking

- Split at headings first, then at paragraphs, then at sentences. Never at a fixed count.
- For a 512-token model: 400 tokens of content plus 50 tokens of overlap, so a 70,000-token filing becomes about 175 chunks.
- Carry the CIK, the accession number and the section heading on every chunk.



Aggregating Chunk Scores

Filing	s1	s2	s3	s4	s5	s6
A	+0.90	-0.70	-0.60	-0.50	-0.40	-0.30
B	+0.30	+0.20	+0.10	+0.10	0.00	-0.10

Rule	Filing A	Filing B	Which looks worse
Mean score	-0.27	+0.10	A
Fraction negative	5 / 6 = 0.83	1 / 6 = 0.17	A
Maximum score	+0.90	+0.30	B

A single sentence reverses the ranking that the other two rules agree on.



Lexical Match and Semantic Match

A company profile built from its own filings: Apple Inc., AAPL, iPhone, App Store, Cupertino, services revenue.

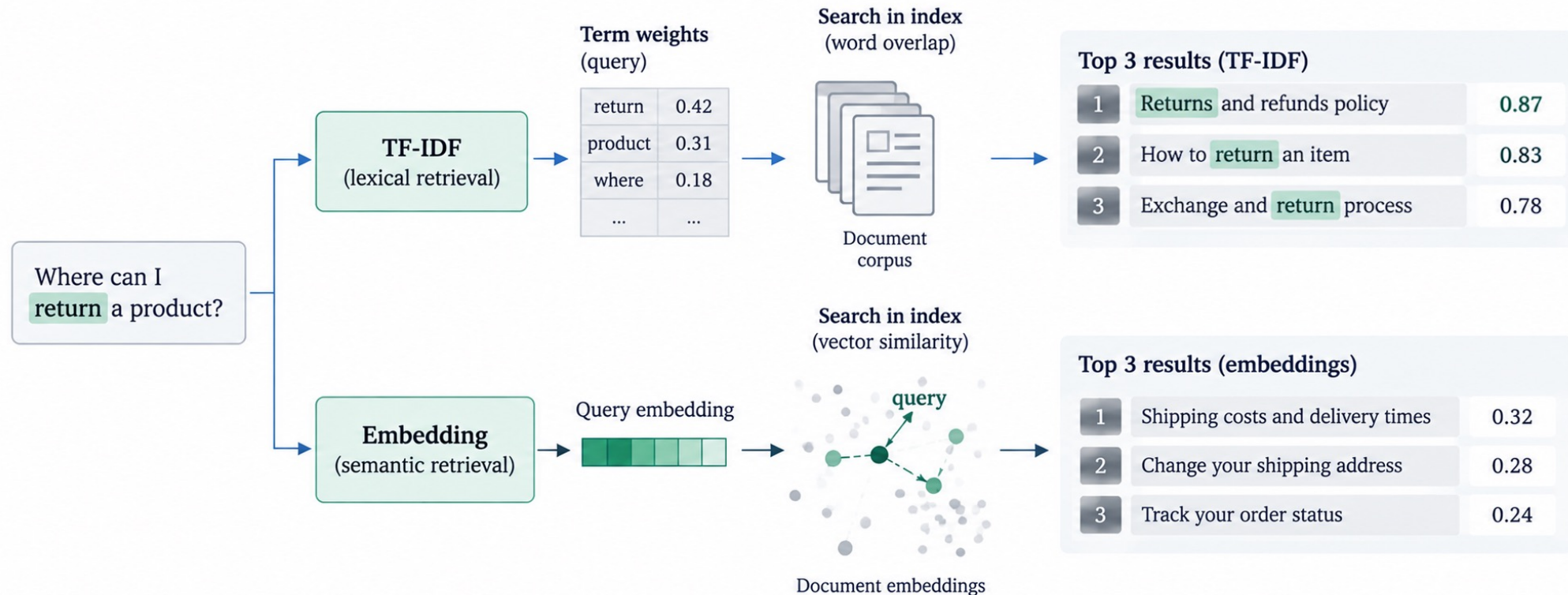
Document	TF-IDF	Embedding	Which is right
"AAPL slipped 2% in late trading."	High	High	Both, correctly
"The Cupertino handset maker warned on shipments."	Near zero	High	The embedding
"Qualcomm and Samsung cut handset chip orders."	Near zero	High	TF-IDF

TF-IDF misses the paraphrase, and the embedding returns the wrong firm.



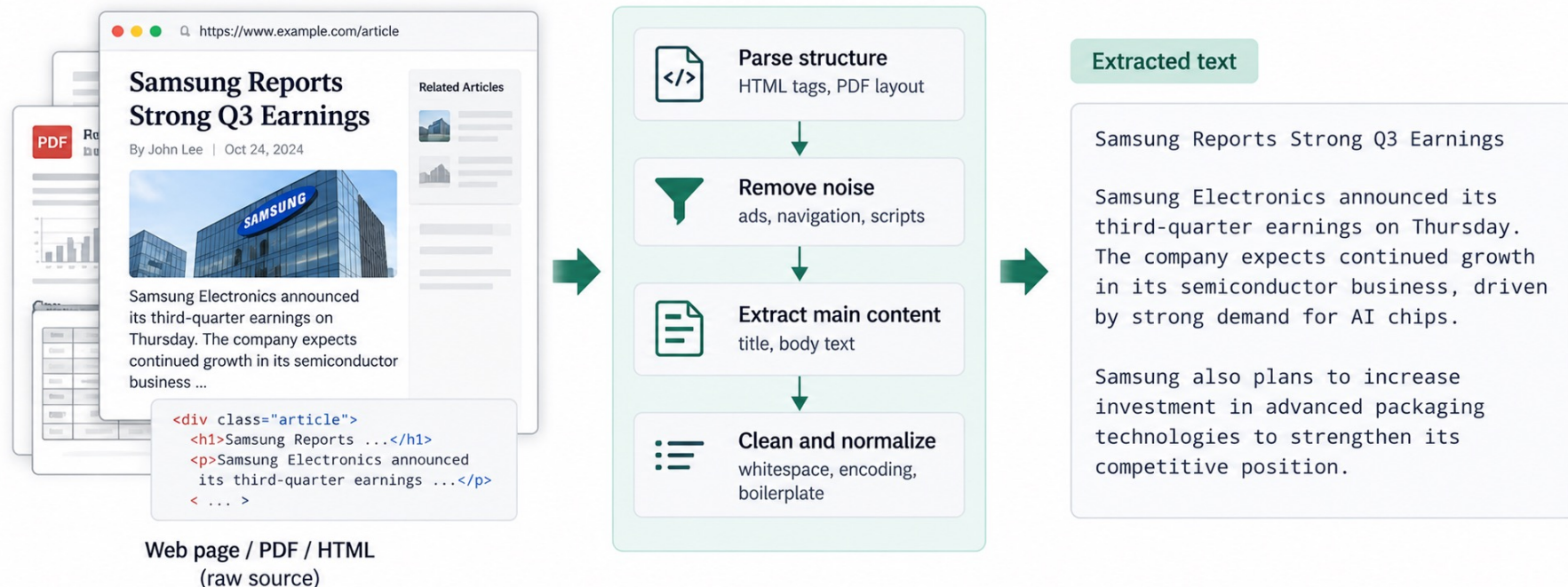
Retrieval With TF-IDF and With Embeddings

- Build a profile per company: concatenate its filings, then fit TF-IDF over all 500 profiles.
- Score an article: run the same vectorizer, take the cosine against all 500, keep whatever clears the threshold.
- Repeat with dense vectors: the paraphrase becomes reachable, and so do the competitors.



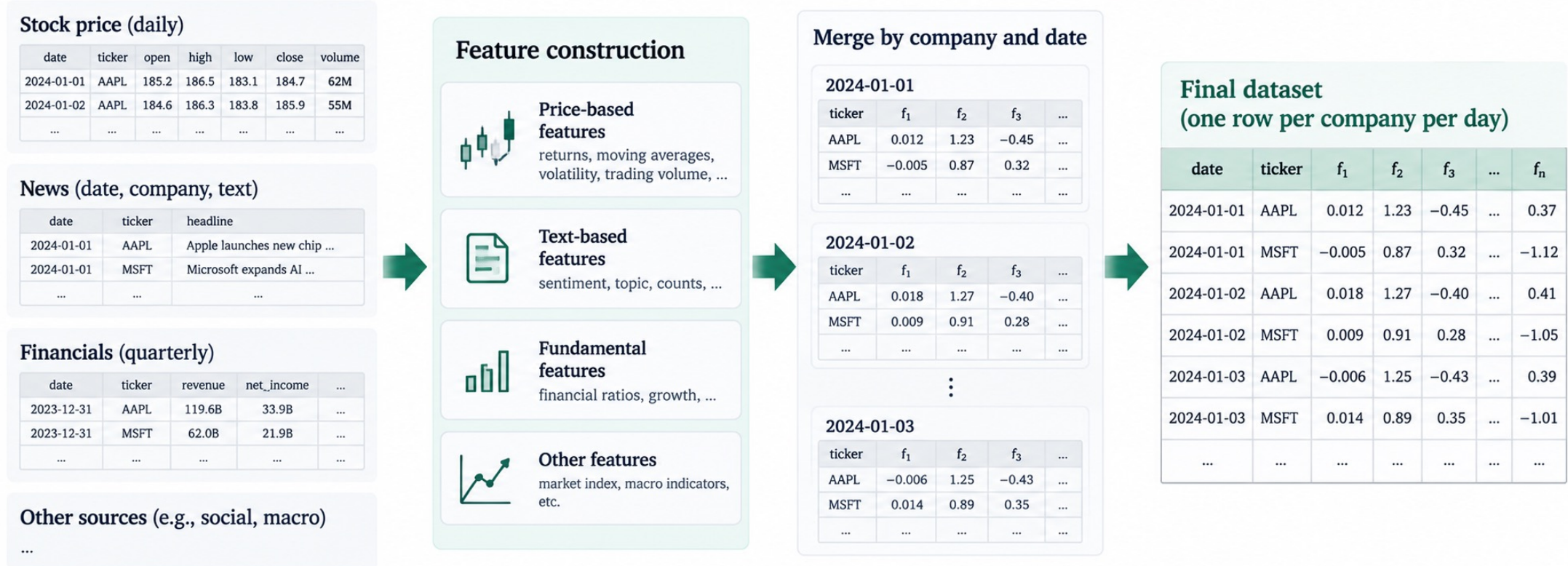
Getting Clean Text Out of a Source

- A web page carries navigation, related links and cookie notices around the article body.
- A PDF returns text in column order, so a two-column page interleaves two sentences.
- Tables lose their row labels once flattened into running text, and the numbers float free.
- A filing package holds a cover page and exhibits, and the exhibit is often the part that matters.



A Daily Feature Per Company

- With identifiers and timestamps attached, the stream becomes a panel of firm-days.
- Per firm per day: the article count, the mean tone, and the dispersion of tone.
- Use the publication time: anything after the close belongs to the next session.



Features From One Day of Reddit Posts

One ticker, one day, one row: six columns built from the same set of posts.

Column	Definition	What it is meant to capture
post_count	Posts mentioning the ticker	Attention
author_count	Distinct authors that day	Whether attention is broad or one person
mean_tone	Average sentiment over posts	Direction
tone_sd	Standard deviation of sentiment	Disagreement between authors
upvote_ratio_mean	Mean upvote ratio of the posts	Whether the board agrees with the posts
new_author_share	Share of posts from accounts under 30 days old	Coordinated posting

Average by author rather than by post, or one active account sets the value of the column.



Features From One Day of News

The same row for a news stream, where the timestamp carries more weight than the wording.

Column	Definition	Why it is separate
article_count	Articles left after near-duplicate removal	A wire story is reprinted many times
first_report_flag	1 for the first article on an event	Later copies carry no new information
headline_tone	Sentiment of the headline only	Headlines are written to be read alone
body_tone	Sentiment of the body text	Usually milder than the headline
source_tier	Wire, national outlet, aggregator	Sets the weight in the average
after_close_flag	1 if published after the close	Assigns the item to the next session

Build these two tables first. A sentiment model only replaces the tone columns.

